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Note

A kernel of order $2k - c$ for Vertex Cover

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ABSTRACT

The best known kernel for the vertex cover problem was of order $2k$ for a long time, by a result of Chen et al. Recently, results by Chlebík and Clebíková, have implied that this can be improved to $2k - 1$. In this paper, we provide a new structural result which can be used to improve this to $2k - c$, for any constant c.

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1. Introduction

Given a graph $G = (V, E)$, with vertex set V and edge set E, a *vertex cover* is a set $X \subseteq V(G)$ such that for every edge $uv \in E$ we have $\{u, v\} \cap X \neq \emptyset$. The size of a minimum vertex cover in G is denoted by $\mathrm{VC}(G)$. In this paper, we will consider *the Vertex Cover Problem* (abbreviated as VCP), which is defined as follows.

The VCP: Given an integer k and a graph G, is $\mathrm{VC}(G) \leq k$?

Vertex cover is one of the fundamental problems in Graph Theory and has many real-world applications. Some of these applications are in the area of biology, where it is related to *Gene Conflict Resolution* and in the analysis of microarray data, to name just a few.

It is easy to produce an approximation algorithm for VCP with approximation ratio 2 (just pick both endpoints of each edge in a maximal matching). However, if the unique games conjecture is true then minimum vertex cover cannot be approximated within any constant factor better than 2 (see [9]).

Using the above definition of the VCP it was well known that there is a *kernel* of VCP of order at most $2k$ (see, Chen et al. [3]). That is, there is a polynomial transformation from any instance of VCP to another one with at most $2k$ vertices, such that the answers to the two problems are equivalent. It was observed in [4] that the results in [4] implied that the $2k$ bound could be improved to $2k - 1$. In this paper, we prove a new structural result and show that the bound can be improved to $2k - c$ for any constant c (where $2k - c > 1$). Our proofs use different ideas and technics to those of [4]. For more information on kernels and Fixed Parameter Tractability, see [5,14].

Chen, Fernau, Kanj and Xia proved in [2] that, unless $NP = P$, the bound cannot be lowered to $(2 - \varepsilon)k$ for any $\varepsilon > 0$, if one requires the kernel to be a subgraph of the original graph. In fact, it is strongly believed that the bound cannot be lowered to $(2 - \varepsilon)k$ even if we allow the kernel not to be a subgraph of the original graph. Therefore it is interesting to improve the bound to $2k - f(k)$ for some (sub-linear) function f of k. In this paper, we make progress in this direction by showing that we may let $f(k)$ be any constant and still find a kernel of order at most $2k - f(k)$.

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Our algorithm to find a kernel of order at most $2k - c$ is purely theoretical, as the time complexity of the algorithm is $O(\sqrt{n}m + k^{5/2} + (k+c)^{c+1})$, which is non-practical unless c is very small. However there is a huge amount of theoretical research going into deciding how small kernels can be, which is where our paper makes its main contribution.

Lemma 1 is the structural result used in our algorithm. Even if our algorithm is not practical when c gets large, the result of Lemma 1 may be useful in future algorithm design.

If $G = (V, E)$ is a graph and $X \subseteq V$ then $G - X$ denotes the graph obtained from G by deleting all vertices in X and all edges with an endpoint in X. The subgraph induced by X will be denoted by $G[X]$ $(=G-(V \setminus X))$. We say that G is connected if there is a path between every pair of vertices. A *component* in G is a maximal connected subgraph. We call a component odd or even depending on the parity of its order.

2. Preliminary results

We first need to define the *fractional vertex cover* of a graph G. Assign a real number $w(x)$ to every vertex x in G, such that $w(x) + w(y) \geq 1$ for all edges $xy \in E(G)$. The minimum possible value of $\sum_{x \in V(G)} w(x)$ is the fractional vertex cover and is denoted by $\mathrm{VC}^f(G)$. It is easy to see that $\mathrm{VC}(G) \geq \mathrm{VC}^f(G)$ and it is well known (and not difficult to prove) that there is an optimal assignment of values to the vertices in G such that all $w(x) \in \{0, 1/2, 1\}$. The following theorem was originally proven by Nemhauser and Trotter in [13] and later different proofs were given by Motwani in [12] and by Khuller in [10].

Theorem 1 ([13]). *There is an $O(\sqrt{n}m)$ time algorithm that, given a graph G of order n and size m, constructs two disjoint subsets V_1 and V_0 of vertices in G such that $\mathrm{VC}(G) = \mathrm{VC}(G[V_0]) + |V_1|$ and $\mathrm{VC}^f(G[V_0]) \geq |V_0|/2$.*

A matching is a set of edges, such that all endpoints are distinct. The following is Edmonds Blossom theorem.

Theorem 2 ([6,11]). *We can find a maximum matching in a graph of order n and size m in time $O(\sqrt{n}m)$.*

2-SAT is the well-known problem of deciding if the conjunction of a set of clauses, each containing the disjunction of two literals, is satisfiable.

Theorem 3 ([7,8]). *There exists an $O(l)$ time algorithm for 2-SAT with l clauses.*

3. A $2k - c$ kernel

The size of a maximum matching in a graph G is denoted by $\alpha'(G)$. Before we prove that there is a kernel of order $2k - c$ for the VCP, we need the following lemma.

Lemma 1. *If G is a graph and V_0 and V_1 are defined as in Theorem 1 then $\mathrm{VC}(G[V_0]) + \alpha'(G[V_0]) \geq |V_0|$.*

Proof. Let G, V_0 and V_1 be defined as in the statement of the lemma and let $G^* = G[V_0]$. According to Berge's theorem [1] there exists a set $X \subseteq V(G^*)$ such that the following holds, where $oc(G^* - X)$ denotes the number of odd components in $G^* - X$.

$$\alpha'(G^*) = \frac{1}{2}(|V(G^*)| + |X| - oc(G^* - X)). \tag{1}$$

Let C be any vertex cover of G^* with $|C| = \mathrm{VC}(G^*)$ and partition all the components of $G^* - X$ into the four sets $Y = \{Y_1, Y_2, \ldots, Y_a\}$, $Z = \{Z_1, Z_2, \ldots, Z_b\}$, $Q = \{Q_1, Q_2, \ldots, Q_{|Q|}\}$ and $P = \{P_1, P_2, \ldots, P_{|P|}\}$, such that all components in Y and Z are odd and all components in Q and P are even and the following holds, where $V(Y)$, $V(Z)$, $V(Q)$ and $V(P)$ denote the vertices in the components in the corresponding set.

$$\forall Y_i \in Y : |C \cap V(Y_i)| > \frac{|V(Y_i)|}{2} \qquad \forall Z_i \in Z : |C \cap V(Z_i)| \leq \frac{|V(Z_i)|}{2}$$
$$\forall Q_i \in Q : |C \cap V(Q_i)| > \frac{|V(Q_i)|}{2} \qquad \forall P_i \in P : |C \cap V(P_i)| \leq \frac{|V(P_i)|}{2}.$$

Let $X' = X \cap C$ and let $X'' = X \setminus C$ and define the following weights for all $v \in V(G^*)$.

$$W(v) = \begin{cases} 1 & \text{if } v \in X' \cup (C \cap V(Z)) \cup (C \cap V(P)) \\ \frac{1}{2} & \text{if } v \in X'' \cup V(Y) \cup V(Q) \\ 0 & \text{if } v \in (V(Z) \setminus C) \cup (V(P) \setminus C)\,. \end{cases}$$

It is not difficult to show that for every edge $uw \in E(G^*)$ we have $W(u) + W(w) \geq 1$. By Theorem 1 we note that $\mathrm{VC}^f(G^*) \geq |V_0|/2 = |V(G^*)|/2$, which by the above implies the following.

$$\frac{|V(G^*)|}{2} \leq \sum_{v \in V(G^*)} W(v) = |X'| + |C \cap V(Z)| + |C \cap V(P)| + \frac{|X''|}{2} + \frac{|V(Y)|}{2} + \frac{|V(Q)|}{2}. \tag{2}$$

As every $Z_i \in Z$ is odd we note that $|C \cap V(Z_i)| \leq \frac{|V(Z_i)|-1}{2}$, which implies that $|C \cap V(Z)| \leq \frac{|V(Z)|-b}{2}$, where we recall that $|Z| = b$. As $|C \cap V(P)| \leq \frac{|V(P)|}{2}$ and $|V(G^*)| = |X'| + |X''| + |V(Y)| + |Y(Z)| + |V(Q)| + |V(P)|$ the above implies the following.

$$\frac{|V(Z)|}{2} + \frac{|V(P)|}{2} \leq \frac{|X'|}{2} + |C \cap V(Z)| + |C \cap V(P)| \leq \frac{|X'|}{2} + \frac{|V(Z)| - b}{2} + \frac{|V(P)|}{2}.$$

Therefore $|X'| \geq b$. We will now show that $|X''| \geq a - i$, where $i = 2|C| - |V(G^*)|$. Note that the following holds.

$$\frac{|V(G^*)|}{2} = |C| - \frac{\mathrm{i}}{2} = |X'| + |C \cap V(Y)| + |C \cap V(Z)| + |C \cap V(Q)| + |C \cap V(P)| - \frac{\mathrm{i}}{2}. \tag{3}$$

Eqs. (2) and (3) now imply the following.

$$|C \cap V(Y)| + |C \cap V(Q)| - \frac{\mathrm{i}}{2} \leq \frac{|X''|}{2} + \frac{|V(Y)|}{2} + \frac{|V(Q)|}{2}. \tag{4}$$

By the definition of Y and Q we have $|C \cap V(Y)| \geq \frac{|V(Y)|+a}{2}$ and $|C \cap V(Q)| \geq \frac{|V(Q)|}{2}$, which by Eq. (4) implies that $|X''| \geq a - i = a + |V(G^*)| - 2|C|$. This was the desired result, which together with $|X'| \geq b$ implies that $|X| \geq oc(G^* - X) + |V(G^*)| - 2|C|$, as $oc(G^* - X) = a + b$. By Eq. (1) we note that the following holds, which completes the proof.

$$\alpha'(G^*) = \frac{1}{2}(|V(G^*)| + |X| - oc(G^* - X)) \geq \frac{1}{2}(2|V(G^*)| - 2|C|) = |V(G^*)| - |C|. \quad \square$$

We are now ready to prove our main result, which is that we can find a kernel of the VCP of order $2k - c$ (in polynomial time), for any constant c.

Theorem 4. *Given integers k and c and a graph G, then there exists an $O(\sqrt{n}m + k^{5/2} + (k+c)^{c+1})$ algorithm that either decides if $\mathrm{VC}(G) \leq k$ or produces a graph G' and an integer $k' \leq k$, such that $\mathrm{VC}(G) \leq k$ if and only if $\mathrm{VC}(G') \leq k'$. Furthermore, $|V(G')| \leq 2k' - c \leq 2k - c$.*

Proof. Let V_0 and V_1 be defined as in Theorem 1 and let $G^* = G[V_0]$. Let $k' = k - |V_1|$ and note that if $|V(G^*)| \leq 2k' - c$ then G^* is the desired kernel. Therefore we may assume that $|V(G^*)| > 2k' - c$.

Let M be any maximum matching in G^* and assume that $M = \{u_1v_1, u_2v_2, \ldots, u_{|M|}v_{|M|}\}$. If $2|M| \leq |V(G^*)| - c$, then by Lemma 1 and $|V(G^*)| > 2k' - c$ we note that $\mathrm{VC}(G^*) \geq |V(G^*)| - |M| \geq \frac{|V(G^*)|+c}{2} > k'$, and therefore there is no solution to the original vertex cover problem. So we may assume that $2|M| > |V(G^*)| - c > 2k' - 2c$. Furthermore, we may assume that $|M| \leq k'$ since otherwise there is also no solution to the original vertex cover problem. Let $X = V(G^*) \setminus V(M) = \{x_1, x_2, \ldots, x_{|X|}\}$.

Our algorithm now looks at all pairs (S_M, S_X) of sets, such that $S_M \subseteq M$ and $S_X \subseteq X$ and $|S_M| + |S_X| + |M| \leq k'$. For each such pair it will determine if there is a vertex cover, C, of G^* such that $C \cap X = S_X$ and S_M are exactly the edges of M containing two vertices from C. We first show how to determine whether such a C exists. If there exists an edge e of G with both endpoints in $X \setminus S_X$, then C does not exist, so assume that this is not the case. Assume, without loss of generality, that $M' = M \setminus S_M = \{u_1v_1, u_2v_2, \ldots, u_lv_l\}$ and build an instance of 2-SAT on the variables $z_1, z_2, \ldots, z_l$ as follows. For each edge e in G^* with $|V(e) \cap V(M')| = 2$ add the following clause.

If $e = u_iu_j$, then add $(z_i \vee z_j)$ If $e = u_iv_j$, then add $(z_i \vee \bar{z}_j)$
If $e = v_iu_j$, then add $(\bar{z}_i \vee z_j)$ If $e = v_iv_j$, then add $(\bar{z}_i \vee \bar{z}_j)$.

Furthermore, for each edge e in G^* with $|V(e) \cap V(M')| = 1$ and $|V(e) \cap (X \setminus S_X)| = 1$ set z_i to be true (or add the clause $(z_i \vee z_i)$) if $e = xu_i$, for some i, and set z_i to false (or add $(\bar{z}_i \vee \bar{z}_i)$) if $e = xv_i$, for some i.

If the resulting instance of 2-SAT is satisfied, then we get the desired vertex cover, C, by adding u_i to C if z_i is true and adding v_i to C if it is false. Furthermore, $S_X \cup V(S_M)$ is added to C. It is not difficult to check that C is a vertex cover and $|C| = |S_X| + |V(S_M)| + l = |S_X| + 2|S_M| + (|M| - |S_M|) \leq k'$.

On the other hand, if the desired vertex cover C exists, then it is not difficult to see that the resulting instance of 2-SAT is satisfied (by letting z_i be true if and only if u_i belongs to C). We have now shown how to determine if C exists.

We will now show that the algorithm solves the VCP in polynomial time. Assume that there exists a vertex cover, C, in G^* with $|C| \leq k'$. Let $S_X = C \cap X$ and let S_M contain the edges of M whose endpoints are both in C. Clearly

$|S_M| + |S_X| + |M| = |C| \leq k'$, so C will be found by our algorithm. So if $\mathrm{VC}(G^*) \leq k'$ our algorithm will find the desired vertex cover and if $\mathrm{VC}(G^*) > k'$, then our algorithm will determine so, by not finding any vertex cover (as any vertex cover the algorithm finds has size at most k'). This completes the description of the algorithm.

We will now compute its complexity, so assume that G has order n and size m. We first need to construct V_1 and V_0, which can be done in $O(\sqrt{n}m)$ time, by Theorem 1. We then need to compute a maximum matching in G^*. If $|V(G^*)| > 2k$, then by Theorem 1 there is no vertex cover of size at most k, so we may assume that $|V(G^*)| \leq 2k$, which implies that we can find a maximum matching in $O(\sqrt{k} \times k^2)$ time, by Theorem 2. We then (in the worst case) need to build and solve an instance of 2-SAT for each pair (S_X, S_M). As $2|M| > |V(G^*)| - c$ we note that $|X| = |V(G^*)| - 2|M| < c$ and we already know that $|M| \leq k'$, so the number of pairs is bounded by $(k' + c)^{c-1}$, as $|S_X| + |S_M| \leq k' - |M| < c$. Furthermore, to build and solve one of the 2-SAT instances takes at most $O((k' + c)^2)$ time as this is a bound on the number of edges in G^*. Therefore the complexity is bounded by $O(\sqrt{n}m + k^{5/2} + (k' + c)^2(k' + c)^{c-1}) = O(\sqrt{n}m + k^{5/2} + (k + c)^{c+1})$. □

Corollary 1. *We can find a kernel of order at most $2k - c$ for the VCP with parameter k in time $O(\sqrt{n}m + k^{5/2} + (k + c)^{c+1})$, whenever $2k - c > 1$.*

Proof. By Theorem 4 we can either solve the problem or produce the desired kernel in $O(\sqrt{n}m + k^{5/2} + (k + c)^{c+1})$ time. If we solve the problem then the kernel is a graph with one edge and the parameter is 1 if the answer to the original problem was "YES" and otherwise it is zero. □

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